

SLC5A3-Dependent Myo-inositol Auxotrophy in Acute Myeloid Leukemia



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ABSTRACT

An enhanced requirement for nutrients is a hallmark property of cancer cells. Here, we optimized an *in vivo* genetic screening strategy in acute myeloid leukemia (AML), which led to the identification of the myo-inositol transporter SLC5A3 as a dependency in this disease. We demonstrate that SLC5A3 is essential to support a myo-inositol auxotrophy in AML. The commonality among SLC5A3-dependent AML lines is the transcriptional silencing of *ISYNA1*, which encodes the rate-limiting enzyme for myo-inositol biosynthesis, inositol-3-phosphate synthase 1. We use gain- and loss-of-function experiments to reveal a synthetic lethal genetic interaction between *ISYNA1* and *SLC5A3* in AML, which function redundantly to sustain intracellular myo-inositol. Transcriptional silencing and DNA hypermethylation of *ISYNA1* occur in a recurrent manner in human AML patient samples, in association with *IDH1/IDH2* and *CEBPA* mutations. Our findings reveal myo-inositol as a nutrient dependency in AML caused by the aberrant silencing of a biosynthetic enzyme.

SIGNIFICANCE: We show how epigenetic silencing can provoke a nutrient dependency in AML by exploiting a synthetic lethality relationship between biosynthesis and transport of myo-inositol. Blocking the function of this solute carrier may have therapeutic potential in an epigenetically defined subset of AML.

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INTRODUCTION

A consistent property of cancer cells is metabolic dysfunction, which can occur to support the energetic and biosynthetic requirements of a rapidly growing tumor (1). One manifestation of this process is that tumor cells become more dependent on extracellular nutrient uptake for their growth and viability than do normal tissues, a process referred to as auxotrophy (2). In many tumors, auxotrophies are driven by an elevated anabolic demand for metabolite building blocks, as is the case for serine (3), glutamine (4), and cysteine (5). However, cancer cells can also develop nutrient dependencies through an acquired defect in *de novo* metabolite biosynthesis pathways. One of the clearest examples of this is the

dependence of lymphoid cancers on extracellular asparagine, which occurs because the rate-limiting enzyme in asparagine biosynthesis becomes epigenetically silenced in this malignancy (6). This nutrient dependency can be exploited by infusion of recombinant asparaginase, which is an approved chemotherapeutic in acute lymphoblastic leukemia that depletes asparagine from the plasma (7). Arginine and cholesterol auxotrophies in cancer have also been attributed to diminished expression of key biosynthetic enzymes (8, 9). While these examples point to the therapeutic significance of auxotrophy in human cancer, few systematic surveys of nutrient dependencies have been performed in cancer to date.

Solute carriers (SLC) are an important class of transporter proteins that regulate nutrient utilization in normal and

neoplastic contexts. The human genome encodes over 400 SLC proteins organized into 65 subfamilies, which regulate intracellular levels of amino acids, sugars, ions, lipids, neurotransmitters, and drugs (10). SLCs contain transmembrane alpha helices that form a substrate binding channel, which often use ion gradients to cotransport substrates into cells (11). Genetic targeting of SLCs, owing to their specificity for transporting specific substrates, provides a powerful tool for manipulation of intracellular metabolite levels and for exploration of nutrient dependencies (12, 13). Numerous SLC genes are mutated in human genetic disorders, and an emerging body of evidence suggests a role for SLC proteins in the pathogenesis of human cancer (14–16). Moreover, the substrate binding cavity of SLC proteins can be blocked with small molecules, which has fueled an interest in this class of proteins as drug targets (16–18).

Myo-inositol is an abundant carbocyclic sugar alcohol in eukaryotic cells (19, 20). One of the most well-characterized functions of myo-inositol is as a head group component of lipid phosphatidylinositols (PI), which are important signaling molecules that function downstream of growth factor signaling. In this lipid context, the phosphorylation state of myo-inositol is highly regulated by kinases (e.g., PI3K) and phosphatases (e.g., PTEN) to control the AKT-mTOR signaling axis (21). In addition, phospholipase C can release inositol triphosphates from PIs, which function as a second messenger that promotes the intracellular release of calcium from the endoplasmic reticulum (22, 23). Myo-inositol is also an osmolyte, whose intracellular concentrations can be regulated to allow cells to survive in hypertonic environments (24, 25).

Intracellular myo-inositol can be derived from three different sources: (i) *de novo* biosynthesis from glucose (26, 27), (ii) regeneration from inositol phosphates via the action of phosphatases (28), or (iii) uptake from the extracellular environment by inositol transporters (29). For *de novo* biosynthesis, glucose is first phosphorylated by hexokinase, then converted to myo-inositol-1-phosphate by the enzyme inositol-3-phosphate synthase 1 (ISYNA1), and finally dephosphorylated by inositol monophosphatase to form free myo-inositol. Importantly, ISYNA1 catalyzes the rate-limiting step in myo-inositol biosynthesis (30). For extracellular uptake, there are three known myo-inositol SLC transporters in humans: SLC5A3 (31), SLC5A11 (32), and SLC2A13 (33). Both SLC5A3 and SLC5A11 are sodium ion-coupled inositol transporters, whereas SLC2A13 is a proton-coupled inositol transporter (29). Among these transporters, SLC5A3 is the most widely expressed across different tissues and has the highest affinity for myo-inositol (33–37).

In this study, we set out to identify genetic dependencies needed for the growth of acute myeloid leukemia (AML) cells *in vivo*. This approach identified the myo-inositol transporter SLC5A3 as an *in vivo*-relevant dependency unique to AML. We show that this dependency is caused by aberrant silencing of *ISYNA1*. While loss of *ISYNA1* has no detectable effect on the fitness of AML cells, it causes an enhanced dependency on SLC5A3 and on extracellular myo-inositol for cell viability. Taken together, our findings reveal myo-inositol auxotrophy as a metabolic vulnerability in AML.

RESULTS

A Domain-Focused CRISPR Screening Method for Identifying *In Vivo*-Relevant AML Dependencies

The genetic dependencies of a cancer cell can be markedly altered by *in vitro* versus *in vivo* growth conditions (38), particularly the requirement for cell-surface receptors and metabolic enzymes whose function is influenced by the extracellular microenvironment. This issue motivated us to develop a robust screening method for interrogating genetic dependencies of human AML cells grown *in vivo*. For this study, we employed our previously cloned domain-focused single-guide RNA (sgRNA) libraries targeting transcription factors (39), chromatin regulators (40, 41), and kinases (42). In addition, we cloned domain-focused sgRNA libraries targeting G-protein-coupled receptors, ion channels, SLCs, glycosyltransferases, proteases, and membrane proteins, which were all cloned into the LRG2.1T vector backbone that contains an optimized sgRNA2.1 scaffold (43). Our screens were performed by lentiviral transduction of each sgRNA library into the AML cell line MOLM-13, engineered to express caspase 9 (Cas9) and firefly luciferase. A key optimization step was to select for high engraftment efficiency of this MOLM-13 line through pretransplantation before running the pooled screen (Fig. 1A). After transduction with an sgRNA library, MOLM-13 cells were injected via tail vein into 10 to 20 immune-deficient mice. MOLM-13 cells were allowed to expand for 12 days *in vivo*, followed by the mice being sacrificed and genomic DNA being isolated from bone marrow and spleen tissue for deep sequencing of the PCR-amplified sgRNA cassette (Fig. 1A). While establishing this procedure, we found that pooling of bone marrow or spleen samples prior to PCR maximized sgRNA representation and the overall accuracy of the screens. Importantly, the sgRNA read count distribution (a reflection of screen quality) and the depletion pattern of control sgRNAs of *in vivo* samples were comparable with the same screen being performed *in vitro* (Supplementary Fig. S1; Supplementary Table S1). We have previously characterized AML dependencies *in vitro* using sgRNA libraries targeting kinases (42), transcription factors (39), and chromatin regulators (40, 41). The *in vivo* CRISPR screening showed highly consistent results with our previous findings (Fig. 1B–D). The *in vivo* screen results between pooled spleen and bone marrow samples were consistent as well (Supplementary Fig. S2). The overall accuracy of our screen was further supported by the *in vivo*-specific depletion of sgRNAs targeting *CD47* (44–46) and *CXCR4* (47), which are known AML dependencies that regulate interactions with the *in vivo* microenvironment (Fig. 1E). In contrast to previous *in vivo* RNA interference screening in leukemia that showed only approximately 10% of *in vivo* dependencies overlapping with *in vitro* targets (48), our screening showed largely consistent results between *in vivo* and *in vitro* conditions (Fig. 1F; Supplementary Fig. S3; Supplementary Table S1). From this screening effort targeting a total of 5,768 genes, we identified 239 targets as *in vivo*-relevant dependencies (Fig. 1F; Supplementary Table S1).

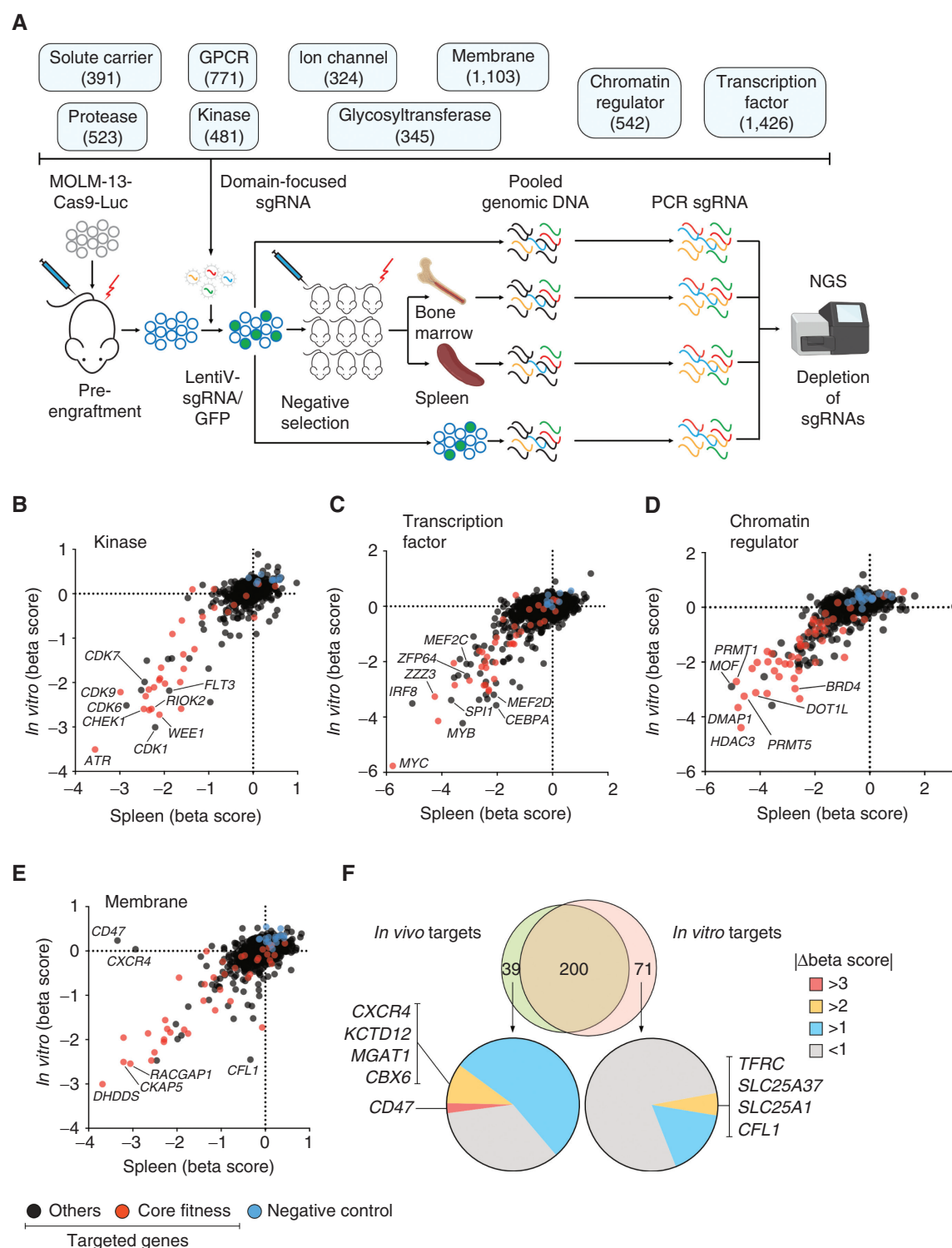


Figure 1. A domain-focused CRISPR screening method for identifying *in vivo*-relevant AML dependencies. **A**, Overview of *in vivo* genetic screening procedure using domain-focused sgRNA libraries. GPCR, G-protein-coupled receptor; NGS, next-generation sequencing. The numbers represent the number of genes in each sgRNA library. Images for mouse bone marrow, spleen, and NGS were created with a licensed version of BioRender. **B–E**, Examples of *in vivo* CRISPR screening results with sgRNA libraries targeting kinases (**B**), transcription factors (**C**), chromatin regulators (**D**), and membrane proteins (**E**). The gene beta score was calculated using MAGeCK (78, 79). The beta score describes how the gene knockout alters cell fitness: A positive beta score indicates positive selection, and a negative beta score indicates negative selection. The screening results using pooled spleen samples were compared with *in vitro* screening results. Core fitness genes (95) are in red; negative control sgRNAs are in blue. **F**, Venn diagram comparing genetic dependencies identified by *in vivo* screening with *in vitro* screening. The *in vivo*-relevant targets were determined using spleen data with $P < 0.05$, FDR < 0.25 , and beta score < 0 . For *in vivo*- and *in vitro*-specific targets, the absolute values of differential beta scores are shown as pie charts. Genes with a differential beta score of 2 or greater are listed.

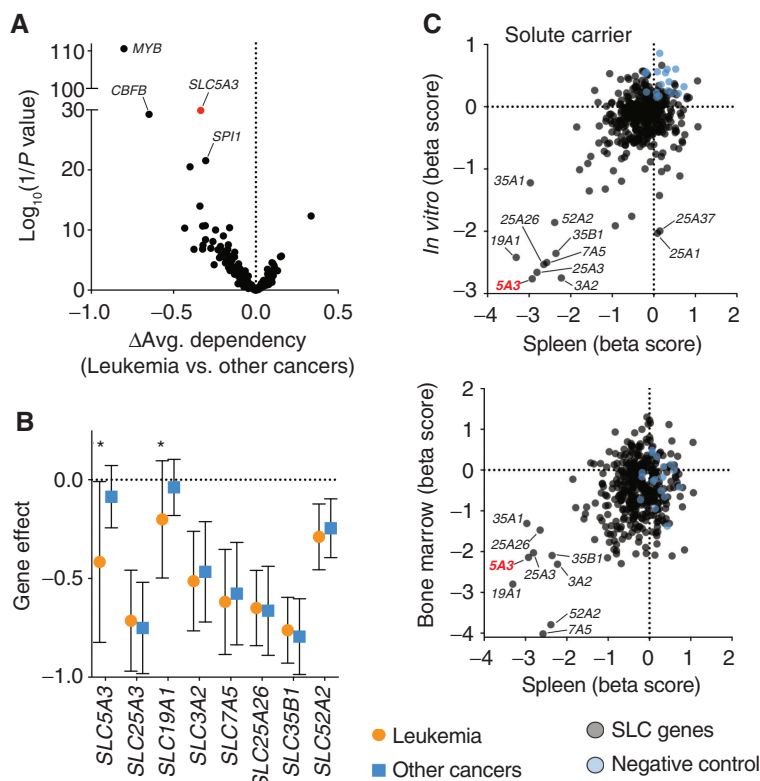


Figure 2. Dependency on SLC5A3 is *in vivo*-relevant and AML-specific. **A**, Analysis of Project Achilles genetic screening data, evaluating the leukemia-biased essentiality of 239 *in vivo*-relevant dependencies. The *in vivo* screening dependencies were determined by $P < 0.05$, FDR < 0.25 , and beta score < 0 . The average gene effect scores were calculated using Achilles Gene Effect 20Q3 data. The P values compared gene effect scores in leukemia versus other cancers using an unpaired Student t test. **B**, Comparison of dependencies of top SLC candidates between leukemia and other types of cancers. The dependency of SLC5A3 is the most leukemia-biased among the candidates. Data are obtained from DepMap database (20Q3). Plotted as average gene effect score with SD in leukemia cell lines versus nonleukemia cell lines. The P values were calculated using unpaired Student t test (*, $P < 0.05$). **C**, Summary of *in vivo* versus *in vitro* CRISPR screening results of the SLC sgRNA library.

SLC5A3 Dependency Is *In Vivo*-Relevant and AML-Specific

While our original intent was to identify *in vivo*-specific dependencies, the scarcity of such hits in our screen prompted us to explore instead whether any *in vivo*-validated dependencies were unique to AML relative to other cancers. For this purpose, we turned to Project Achilles/DepMap, which has performed genome-wide CRISPR screens for essential genes in more than 769 diverse cancer cell lines grown *in vitro* (49, 50). Through analysis of these data, we nominated myeloid lineage TFs (MYB, CBFB, and SPI1) and the myo-inositol transporter protein SLC5A3 as the top leukemia-biased dependencies among the pool of *in vivo*-validated targets (Fig. 2A). The AML specificity of SLC5A3 was distinct from other essential SLC genes, which tended to be needed across diverse cancer types (Fig. 2B). The *in vivo* screening in MOLM-13 cells and our own *in vitro* SLC screening in 10 diverse cancer cell lines corroborated SLC5A3 as a unique AML dependency under *in vitro* and *in vivo* conditions (Fig. 2C; Supplementary Fig. S4A–S4E). Unlike the aforementioned myeloid TFs, no prior study has investigated the role for SLC5A3 in AML, which motivated our subsequent evaluation of this target.

Validation of SLC5A3 Dependency in a Subset of AML Cell Lines

To validate SLC5A3 as a dependency in AML, we performed competition-based proliferation assays following lentiviral transduction with sgRNA-expressing vectors in Cas9-expressing AML cell lines. The genome editing efficiency of these sgRNAs targeting SLC5A3 was validated using genomic DNA qPCR with primers that hybridize to the Cas9 target sequence (Supplementary Fig. S5A). Using this approach, we validated that the cell fitness of several AML lines was impaired by SLC5A3 sgRNAs (e.g., MOLM-13, SEM, MV4-11, U937, EOL1), whereas other AML lines were SLC5A3 independent (e.g., HEL, KASUMI, SET2; Fig. 3A; Supplementary Fig. S5B and S5C). As a control, an sgRNA targeting the DNA replication protein PCNA suppressed the growth of all cancer cell lines tested (Fig. 3A; Supplementary Fig. S5B). cDNA rescue experiments also supported that these cell fitness alterations were due to a requirement for the SLC5A3 gene product and ruled out off-target effects as contributing to this phenotype (Supplementary Fig. S6A–S6E). In MOLM-13 cells, loss of SLC5A3 led to cell-cycle arrest (Supplementary Fig. S7A and S7B). We also verified that knockout of SLC5A3 suppressed the growth of MOLM-13 cells under *in vivo* conditions

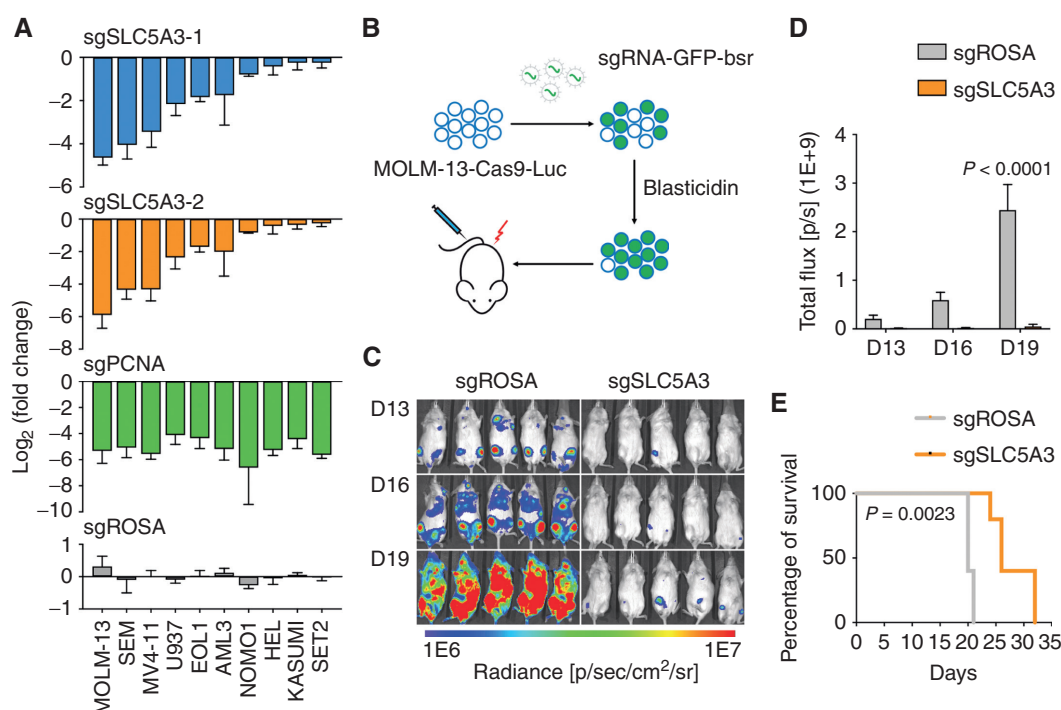


Figure 3. Validation of SLC5A3 dependency. **A**, Summary of *in vitro* validation experiments of individual sgRNAs (linked to GFP) in competition-based proliferation assays in AML cell lines. Plotted is the log₂-transformed fold change of GFP-positive cell percentage at day 18 versus day 3 after infection of sgRNA lentivirus. Shown as mean with SD of two to seven biological replicates. **B**, Procedure of *in vivo* validation of SLC5A3 dependency in human AML. MOLM-13 cells expressing Cas9 and luciferase were transduced with sgROSA or sgSLC5A3-2 (linked to GFP/bsr) and then selected with blasticidin. The SLC5A3 knock-out cells (sgSLC5A3) and control cells (sgROSA) were then transplanted into NSG mice via tail vein injection. **C**, Bioluminescence imaging of NSG mice transplanted with MOLM-13 cells transduced with the indicated sgRNA. The images were taken on days 13, 16, and 19 posttransplantation. **D**, Quantification of bioluminescence from **C**. Values represent total flux as photons per second (mean $\pm$ SD). The *P* value was calculated using unpaired Student *t* test (*n* = 5). **E**, Survival curves of mice shown in **C**. The *P* value was calculated using log-rank (Mantel-Cox) test (*n* = 5).

(Fig. 3B–E; Supplementary Fig. S8A–S8F). Collectively, these findings demonstrate that SLC5A3 is essential in a subset of AML cell lines under *in vitro* and *in vivo* conditions.

SLC5A3-Dependent AML Lines Are Myo-inositol Auxotrophs

Since the primary substrate for SLC5A3 is myo-inositol (29), we performed LC/MS to measure intracellular myo-inositol levels in cancer cell lines following SLC5A3 inactivation. As expected, knockout of SLC5A3 in MOLM-13 or MV4-11 cells led to a significant reduction of intracellular myo-inositol, whereas the level of other metabolites (e.g., glutamine) was unaffected (Fig. 4A). This finding prompted us to evaluate the impact of removing myo-inositol from the media of AML cell lines. Remarkably, SLC5A3-dependent cell lines MOLM-13, MV4-11, and U937 slowed their growth following myo-inositol depletion, whereas the growth of SLC5A3-independent lines HEL, SET2, and K562 was unaffected by myo-inositol depletion (Fig. 4B and C). Taken together, these experiments suggest that SLC5A3 dependency is associated with a myo-inositol auxotrophy in AML.

It has been shown that developmental abnormalities in *Slc5a3*^{−/−} mice can be rescued by providing supraphysiologic concentrations of myo-inositol in the diet, which can be transported into cells in a SLC5A3-independent manner via

low-affinity transporters (51–53). This observation motivated us to perform analogous experiments in SLC5A3-dependent AML cell lines by supplementing normal growth media with additional myo-inositol. This revealed that an approximately fivefold increase in myo-inositol concentrations in regular growth media (RPMI) was sufficient to bypass the necessity of SLC5A3 in MOLM-13 and MV4-11 cell line contexts (Fig. 4D). These findings lend strong support that the essential function of SLC5A3 in AML is the transport of extracellular myo-inositol into the cell interior.

Transcriptional Silencing of ISYNA1 Leads to SLC5A3 Dependency through a Synthetic Lethal Genetic Interaction

We next investigated the mechanism underlying SLC5A3 dependency and myo-inositol auxotrophy in AML. Importantly, SLC5A3 dependency did not correlate with expression of SLC5A3 or silencing of the alternative myo-inositol transporters SLC5A11 and SLC2A13 (Supplementary Fig. S9A–S9C). Furthermore, triple-knockout experiments targeting SLC5A3, SLC5A11, and SLC2A13, failed to reveal evidence of redundancy (Supplementary Fig. S10A and S10B). These results led us to consider whether the myo-inositol auxotrophy in AML was due instead to an acquired defect in *de novo* myo-inositol biosynthesis (Fig. 5A). By analyzing the

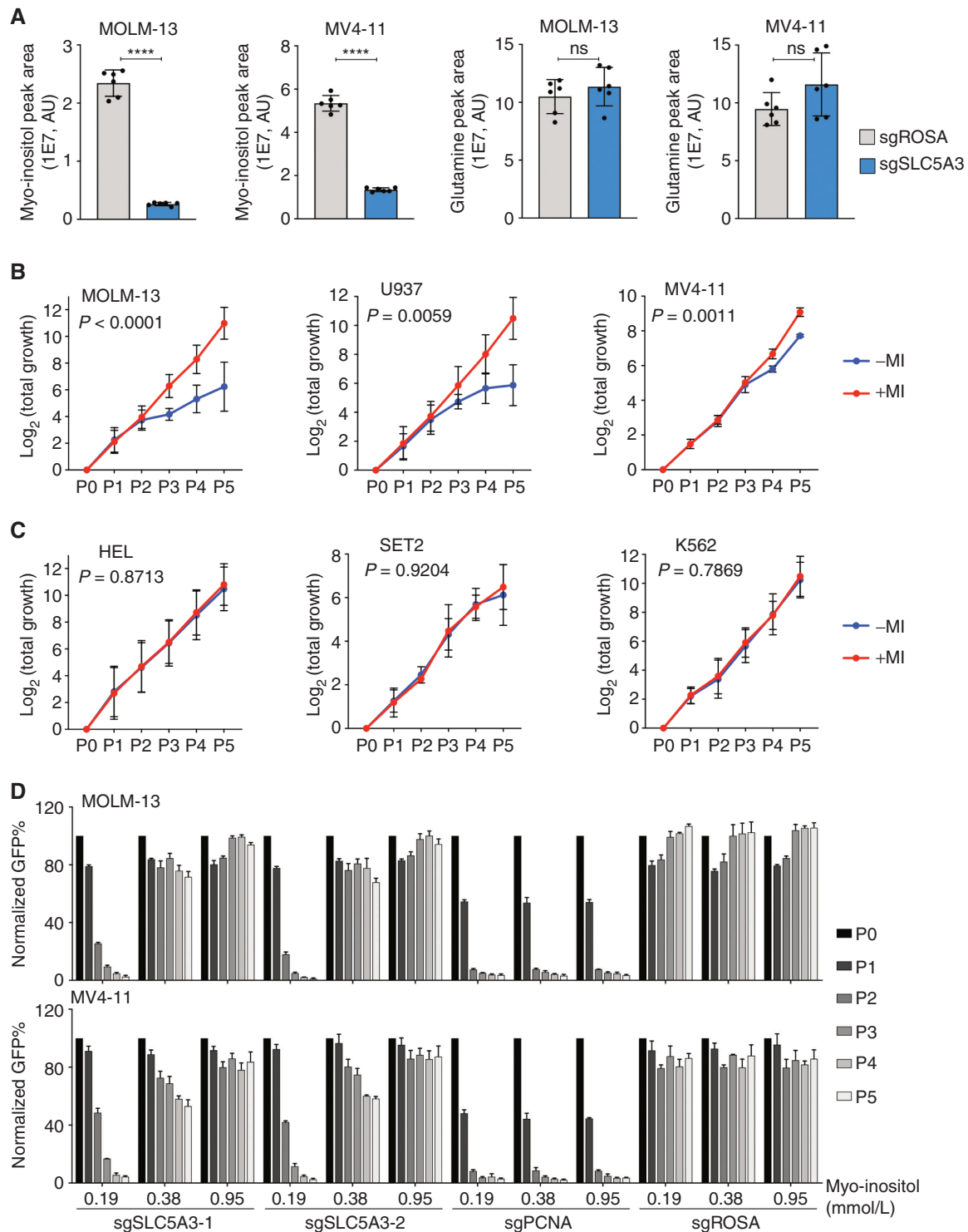


Figure 4. SLC5A3-dependent AML lines are myo-inositol auxotrophs. **A**, LC/MS analysis of cellular myo-inositol and glutamine levels in MOLM-13 and MV4-11 cells transduced with the indicated sgRNA. The P value was calculated using unpaired Student t test ($n = 6$; ****, $P < 0.0001$). ns, not significant. **B** and **C**, Depletion of myo-inositol in the cell culture medium affects the growth of SLC5A3-dependent lines MOLM-13, U937, and MV4-11 (**B**), but not SLC5A3-independent lines HEL, SET2, and K562 (**C**). A customized RPMI cell culture medium without myo-inositol, supplied with dialyzed FBS, was used (-MI). Additional myo-inositol supplement (1.11 mmol/L/0.2 g/L) was added into the medium for rescue (+MI). (Shown as mean with SD of two to four biological replicates; the P values were calculated using two-way ANOVA Sidak test to compare each line graph). **D**, Competition-based proliferation assays in MOLM-13 and MV4-11 cells transduced with the indicated sgRNA and treated with additional myo-inositol supplement. The base level of myo-inositol in RPMI medium is 0.035 g/L (0.19 mmol/L). To rescue the phenotype, additional myo-inositol was added to the cell culture to make myo-inositol concentration as 0.07 g/L (0.38 mmol/L) and 0.175 g/L (0.95 mmol/L). Myo-inositol from FBS (100% FBS contains ~0.9 mmol/L, determined by LC/MS, 10% FBS was added to cell culture medium) was not included (shown as mean with SD of three replicates with independent infections).

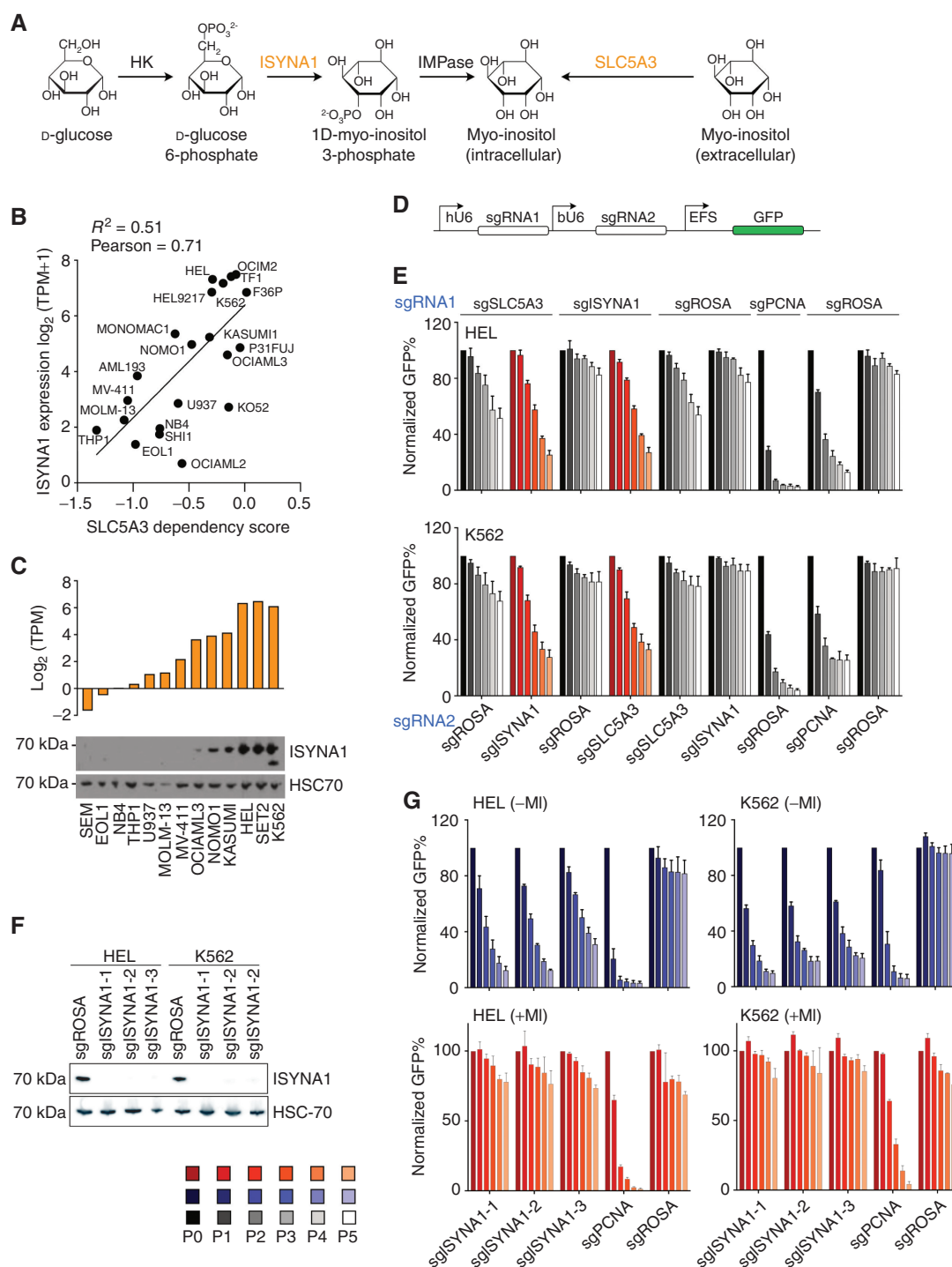


Figure 5. Transcriptional silencing of *ISYNA1* leads to SLC5A3 dependency through a synthetic lethal genetic interaction. **A**, Summary of two major sources of cellular myo-inositol: transported from extracellular environment by SLC5A3 and *de novo* biosynthesis from glucose. The rate-limiting *de novo* myo-inositol biosynthesis enzyme is inositol-3-phosphate synthase 1, encoded by the *ISYNA1* gene. **B**, SLC5A3 dependency in AML (and K562) cell lines correlates with *ISYNA1* expression. Data from DepMap database (Achilles 20Q3). TPM, transcripts per kilobase million. **C**, Expression of *ISYNA1* in AML (and K562) cell lines. The *ISYNA1* gene mRNA expression data were obtained from the Cancer Cell Line Encyclopedia (CCLE) database (96, 97). The *ISYNA1* protein expression in AML lines and K562 (chronic myeloid leukemia) was detected by Western blot analysis, shown as one of two replicates. **D**, A double-knockout construct that cotargets two genes by two sgRNAs driven by the human U6 (hU6) or bovine U6 (bU6) promoter, respectively. The sgRNAs are linked to GFP. **E**, Competition-based proliferation assays in Cas9-expressing SLC5A3-independent cell lines following infection with the indicated cotargeting sgRNA constructs. Double knock-out both *ISYNA1* and SLC5A3 leads to growth phenotype (red; shown as mean with SD of three replicates of independent infections). **F**, Western blot shows depletion of the *ISYNA1* protein in HEL and K562 cells following infection with the indicated sgRNA. **G**, Competition-based proliferation assays in Cas9-expressing HEL and K562 cells following infection with the indicated sgRNA. The experiment was performed in myo-inositol-deficient RPMI medium without myo-inositol supplement (–MI, blue), or with 0.05 g/L (0.28 mmol/L) myo-inositol supplement (+MI, red; shown as mean with SD of three replicates with independent infections).

mRNA levels of known myo-inositol biosynthesis genes, we noticed a significant inverse correlation between *SLC5A3* dependency and expression of *ISYNA1*, which encodes the rate-limiting enzyme in myo-inositol biosynthesis (ref. 30; Fig. 5B). We confirmed the heterogeneous expression of *ISYNA1* in AML cell lines using Western blotting, which mirrored the heterogeneity of mRNA levels (Fig. 5C; Supplementary Fig. S11A).

The findings above led us to hypothesize that a genetic redundancy may exist between *ISYNA1* and *SLC5A3*, which represent two independent strategies for maintaining intracellular myo-inositol levels. To evaluate this possibility, we targeted *ISYNA1* and *SLC5A3* (alone and in combination) in HEL and K562, which are leukemia lines that express both genes. Competition-based proliferation assays using bicistronic sgRNA vectors demonstrated that dual targeting of *ISYNA1* and *SLC5A3* led to a more severe proliferation defect than the loss of each gene individually (Fig. 5D and E; Supplementary Fig. S11B). LC/MS analysis demonstrated that a combined deficiency of *ISYNA1* and *SLC5A3* also led to a more severe decrease in intracellular myo-inositol than observed following either single gene knockout (Supplementary Fig. S11C). Knockout of *ISYNA1* in HEL cells also led to a significant reduction of intracellular myo-inositol when cultured in myo-inositol-deficient media (Supplementary Fig. S11D). In addition, knockout of *ISYNA1* was sufficient to convert HEL and K562 cells into myo-inositol auxotrophs (Fig. 5F and G). Taken together, these experiments validate a genetic redundancy between *ISYNA1* and *SLC5A3* to sustain intracellular myo-inositol and cell viability.

To validate that *ISYNA1* silencing is the underlying cause of myo-inositol auxotrophy and *SLC5A3* dependency in AML, we lentivirally expressed a *ISYNA1* cDNA in AML cell lines that lacked endogenous *ISYNA1* expression (MOLM-13, MV4-11, EOL1, and U937; Fig. 6A). Remarkably, restoring *ISYNA1* was sufficient to elevate myo-inositol levels and bypass the essentiality of *SLC5A3* when cultured in normal growth media (Fig. 6B and C; Supplementary Fig. S12). This finding was further corroborated using a genome-wide screening approach, which revealed *SLC5A3* as the sole genetic dependency rescued by *ISYNA1* cDNA expression (Supplementary Fig. S13A–S13E). Furthermore, forced *ISYNA1* expression eliminated the myo-inositol auxotrophy (Fig. 6D).

To validate that *ISYNA1* is the key enzyme for *de novo* myo-inositol biosynthesis from glucose, we performed a stable isotope tracing experiment by treating the cells with ^{13}C -labeled glucose for 48 hours and measuring intracellular ^{13}C -labeled myo-inositol by LC/MS (Supplementary Fig. S14A). In MOLM-13 cells, the labeled glucose was converted into myo-inositol only when *ISYNA1* was overexpressed (Supplementary Fig. S14B). On the contrary, a significant reduction of intracellular ^{13}C -labeled myo-inositol was observed in HEL cells following *ISYNA1* knockout (Supplementary Fig. S14C).

Aberrant *ISYNA1* Promoter Hypermethylation Occurs Recurrently in Human AML Patient Samples

We next evaluated whether epigenetic silencing of *ISYNA1* occurs in human AML patient samples. In the cell line AML

models, we noticed that transcriptional silencing of *ISYNA1* was associated with DNA hypermethylation and the loss of histone acetylation in the vicinity of its CpG island/promoter (Supplementary Fig. S15A and S15B). Using Nanopore sequencing, we confirmed DNA hypermethylation at the *ISYNA1* gene in MOLM-13 and MV4-11 lines, as well as hypomethylation in the HEL line (Fig. 7A; Supplementary Fig. S15C). Notably, these regions remained unmethylated in normal human hematopoietic stem and progenitor cells (HSPC; ref. 54; Fig. 7A). Having established hypermethylation as a biomarker of *ISYNA1* silencing, we next evaluated whether this genomic region was hypermethylated in human AML patient samples analyzed by The Cancer Genome Atlas (TCGA), in which 170 genetically annotated samples were evaluated using Illumina Infinium 450k DNA methylation assay (55). In this study, we found that *ISYNA1* hypermethylation was present in approximately 20% of human AML samples (Z score > 0.5), whereas other flanking regions lacked heterogeneity in methylation (Fig. 7B; Supplementary Fig. S15D and S15E). Importantly, the presence of hypermethylation correlated with lower *ISYNA1* expression (Fig. 7C; Supplementary Fig. S15F). In addition, we found that *ISYNA1* hypermethylation was significantly enriched in patients with *IDH1/IDH2* or *CEBPA* mutations (Fig. 7B and D). The TCGA AML patient gene expression data further show that patients with *IDH2* mutations have significantly lower *ISYNA1* gene expression (Fig. 7E). We further evaluated *ISYNA1* methylation in an independent cohort of 119 AML patient samples analyzed using reduced representation bisulfite sequencing (56), which likewise identified hypermethylation in the *ISYNA1* genic region in a subset of AML cases, which was absent in normal bone marrow cells (Supplementary Fig. S16). Taken together, these findings validate that aberrant hypermethylation and silencing of *ISYNA1* occurs recurrently in AML.

DISCUSSION

Our study identifies myo-inositol auxotrophy as a previously unrecognized metabolic vulnerability in AML. We trace this auxotrophy to aberrant silencing of *ISYNA1*, which encodes the rate-limiting enzyme for myo-inositol biosynthesis. Our work suggests that normal cells maintain intracellular myo-inositol levels through the redundant mechanisms of *de novo* biosynthesis and *SLC5A3*-dependent transport from the extracellular microenvironment. Our work shows how *ISYNA1* silencing eliminates this redundancy, which in turn drives an elevated demand for *SLC5A3*-dependent myo-inositol transport in AML (Fig. 7F).

The evolution of redundant mechanisms for sustaining intracellular myo-inositol is likely to reflect the vital function of this metabolite, which is well supported by experiments in diverse eukaryotic species (57). Myo-inositol is incorporated as a head group into a variety of lipids that function to promote signal transduction in growth-promoting pathways (e.g., AKT-mTOR; ref. 21). Hence, blockade of *SLC5A3* in *ISYNA1*-deficient AML is likely to extinguish AKT-mTOR signaling, which would be expected to cause an acute arrest in cell proliferation. Targeting *SLC5A3* would also be expected to deplete the cell of inositol phosphates, which regulate

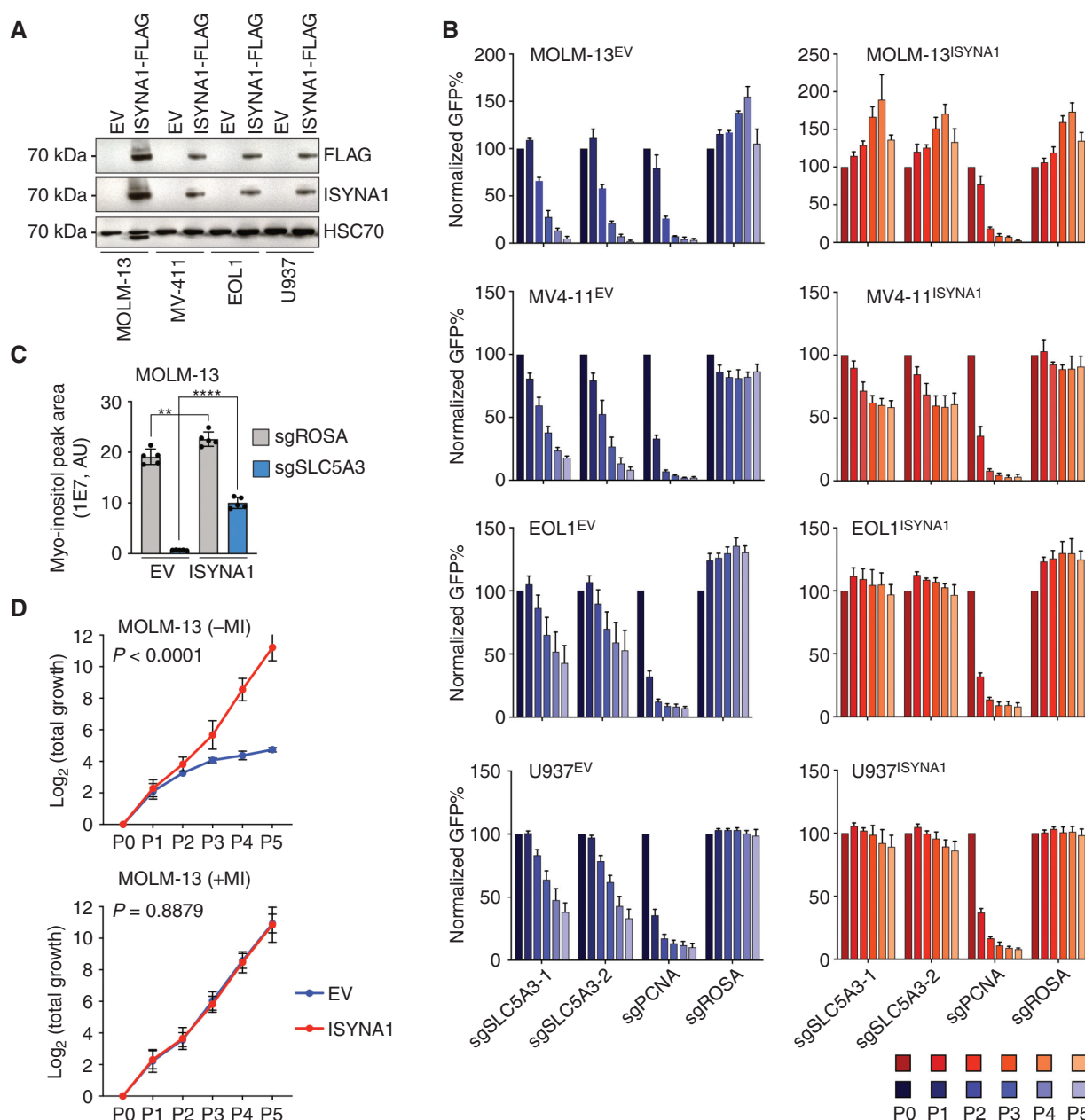


Figure 6. Reexpression of ISYNA1 eliminates SLC5A3 dependency and myo-inositol auxotrophy. **A**, Western blot shows overexpression of FLAG-tagged ISYNA1 in the indicated AML cell lines. **B**, Competition-based proliferation assays in SLC5A3-dependent AML lines expressing empty vector (EV, blue) or ISYNA1 (red), following infection with indicated sgRNA linked to GFP (shown as mean with SD of three replicates from independent infections). **C**, LC/MS analysis in MOLM-13 cells expressing empty vector or ISYNA1 following infection with indicated sgRNA. The *P* value was calculated using two-way ANOVA ($n = 5$; **, $P = 0.0011$; ****, $P < 0.0001$). **D**, Overexpression of ISYNA1 in MOLM-13 cells rescues the growth defect in myo-inositol deficient medium (–MI) but has no impact on the cell growth in medium with myo-inositol supplement (+MI); shown as mean with SD of two biological replicates; each includes three replicates with independent infection; the *P* value was calculated using two-way ANOVA Sidak test).

calcium release as well as the function of chromatin remodeling complexes (57, 58). These myriad functions of myo-inositol for cell viability are likely to contribute to the fitness defects of targeting SLC5A3 in AML.

Our study adds to a growing body of evidence highlighting how epigenetic silencing can alter metabolic pathways and lead to nutrient dependencies. Hypermethylation of the argininosuccinate synthetase (*ASS1*) gene promoter leads to

an arginine dependency in several human cancers (8, 59, 60). Similarly, hypermethylation and silencing of asparagine synthetase (*ASNS*) has been linked to asparagine auxotrophy in lymphoid leukemia (6, 61, 62). Promoter hypermethylation of a rate-limiting cholesterol biosynthetic enzyme gene, squalene mono-oxygenase (*SQLE*), leads to cholesterol auxotrophy in anaplastic large-cell lymphoma (9). It remains unclear as to why the CpG islands of metabolic genes are prone to

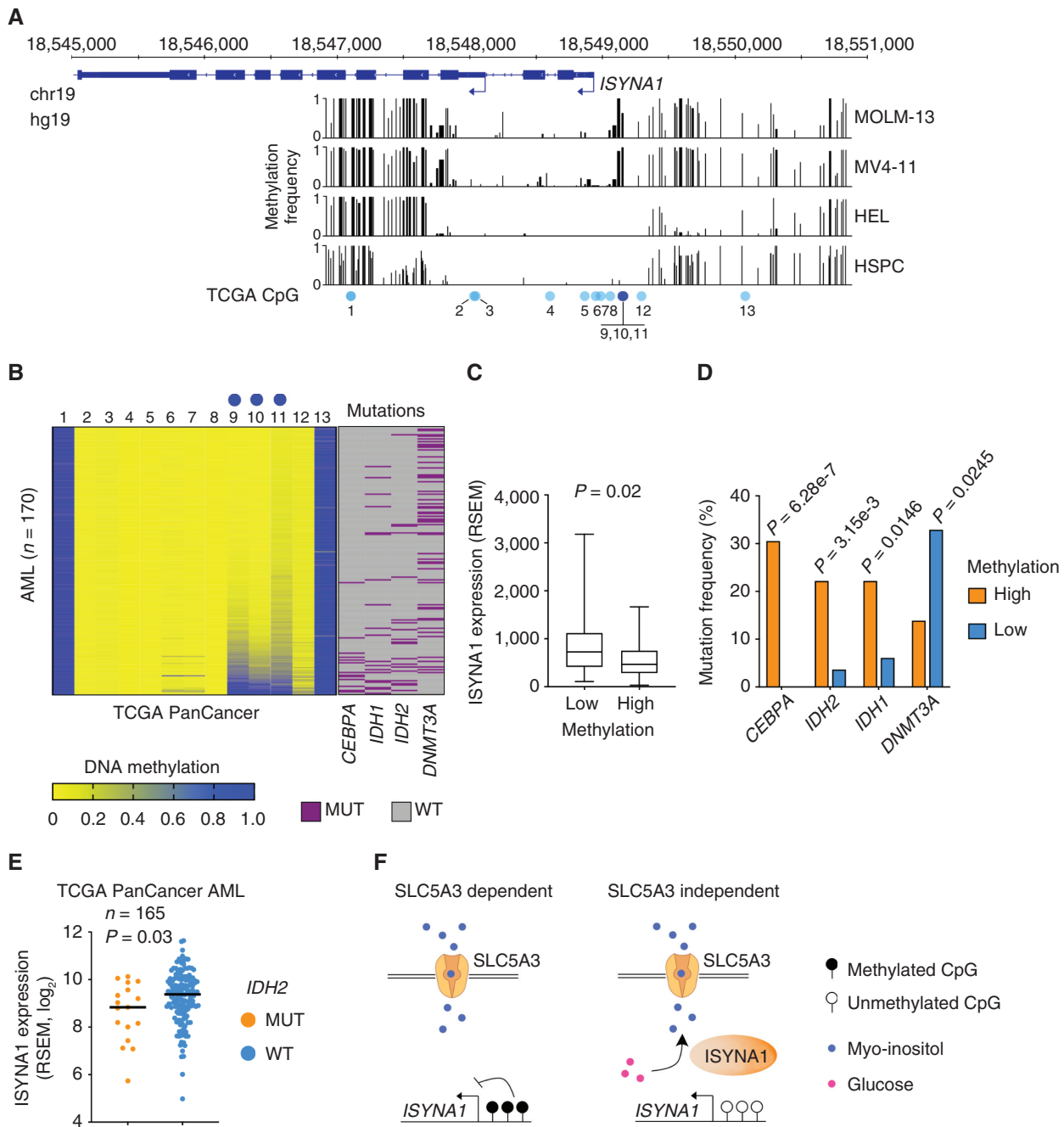


Figure 7. The *ISYNA1* gene is epigenetically silenced by aberrant DNA methylation in a subset of AML. **A** and **B**, CpG methylation status at the *ISYNA1* gene locus in AML cell lines (MOLM-13, MV4-11, HEL) and normal HSPCs (**A**), as well as patients with AML (**B**). The AML cell line methylation was measured by Nanopore sequencing. The HSPC methylation was measured by genome-wide bisulfite sequencing, obtained from a previous study (54). The AML patient data were obtained from the TCGA PanCancer AML database (55). The TCGA AML patient data used a DNA methylation array with reduced CpG representation, shown as “TCGA CpG” (blue dots). The CpGs at the *ISYNA1* promoter (dark blue dots in TCGA CpG) show hypermethylation in SLC5A3-dependent/*ISYNA1*-low AML lines MOLM-13 and MV4-11, but hypomethylated in SLC5A3-independent/*ISYNA1*-high HEL line as well as normal HSPC cells. These regions were also shown aberrantly methylated in a subset of patients with AML (**B**). **C**, *ISYNA1* gene expression level is significantly lower in the high methylation group, compared with the low methylation group ($P = 0.02$). The high and low groups were determined by the Z scores of average DNA methylation levels of the three promoter CpGs (TCGA) described above (Supplementary Fig. S15E). **D**, *CEBPA*, *IDH1*, and *IDH2* mutations are enriched in the high *ISYNA1* methylation group, while *DNMT3A* mutation is enriched in the low methylation group. **E**, Patients with AML with *IDH2* mutations show lower *ISYNA1* gene expression ($P = 0.03$). Data in **D** and **E** were obtained from the TCGA PanCancer AML database via cBioPortal. **F**, A model depicting synthetic lethality between SLC5A3 and *ISYNA1* in AML.

DNA hypermethylation, as it is unclear whether silencing of these genes confers a growth advantage in human cancer cells. Because epigenetic and metabolic regulation is tightly interconnected, it is possible that feedback regulation exists between the chromatin regulatory machinery and metabolite levels. For example, we observe that *ISYNA1*-silenced AML is correlated with the presence of *IDH1/IDH2* gain-of-function mutations, which produce 2-hydroxyglutarate, which in turn inhibits active DNA demethylation (63, 64). Thus, it is possible that the unique metabolic state of AML is the predisposing factor for *ISYNA1* silencing as a passenger epigenetic event in this disease—a possibility that warrants additional investigation.

The aberrant DNA methylation of *ISYNA1* is unlikely to be unique to AML, as several other cancer types (e.g., breast invasion carcinoma) we surveyed in the TCGA database also showed a similar pattern (Supplementary Fig. S17A–S17F). However, the correlation between *SLC5A3* dependency and *ISYNA1* expression is weak in non-AML cell lines. This suggests that other factors might influence myo-inositol levels in non-AML contexts. Nevertheless, these findings raise the possibility that *SLC5A3* dependency and myo-inositol auxotrophy might be present more broadly across human cancers.

Several mechanisms have been described by which the *ISYNA1* gene is regulated in mammalian cells to control myo-inositol biosynthesis, which may predispose to gene silencing in AML. For example, the inositol hexakisphosphate kinase 1 (IP6K1) has been found in the nucleus where it acts directly to maintain transcription of *ISYNA1* and prevent DNA hypermethylation (65). In addition, prior studies have shown that *Musashi2*, a known oncogene, regulates *ISYNA1* expression in the context of cancer (66). The tumor suppressor p53 has also been shown to directly regulate the *ISYNA1* promoter (67). Our own genetic analysis of human samples suggests that mutations of the C/EBP α transcription factor are associated with *ISYNA1* hypermethylation, which may influence the methylation state of this gene. Collectively, these studies suggest that several cancer-relevant pathways converge in regulating *ISYNA1* expression, which might underlie the aberrant silencing of this gene in AML.

Our work raises the possibility that the acquired dependency on *SLC5A3* and extracellular myo-inositol in AML might have therapeutic significance. In analogy to the use of asparaginase in lymphoblastic leukemias (68), one possibility for therapeutic intervention in *ISYNA1*-deficient AML would be to suppress plasma levels of myo-inositol using an infusion of catabolic enzymes. MIOX (myo-inositol oxygenase) is one such enzyme that oxidizes myo-inositol into D-glucuronic acid (69, 70). Another enzyme capable of catabolizing myo-inositol is inositol dehydrogenase from *Bacillus subtilis* (71, 72). As an alternative strategy, several SLCs have been targeted using small molecules (15). Thus, an attractive strategy would be to suppress *SLC5A3* function. While *Slc5a3*-null mice die soon after birth due to respiratory failure from abnormal development of peripheral nerves, this phenotype can be rescued by prenatal myo-inositol supplementation. Adult *Slc5a3*-null mice show a decrease of tissue myo-inositol levels and reduced nerve conduction velocity, which can also be alleviated by myo-inositol supplementation (52). Taken together, our findings justify an investigation of myo-inositol-lowering

agents or *SLC5A3* blockade as therapeutic interventions for the elimination of *ISYNA1*-deficient AML cells.

METHODS

Cell Lines and Culture

All cell lines were authenticated using short tandem repeat profiling. MOLM-13, MV4-11, SEM, OCIAML3, EOL1, U937, NOMO1, HEL, SET2, KASUMI1 (human AML), K562 (human chronic myeloid leukemia, CML), AsPC-1, MIA PaCa-2 (human pancreatic cancer), and NCI-H82 (human small-cell lung cancer, SCLC) cells were cultured in RPMI supplemented with 10% FBS. MA9-ITD and MA9-RAS (engineered human AML) cells were cultured in Iscove's modified Dulbecco's medium supplemented with 20% FBS (73, 74). RH30, RD (human rhabdomyosarcoma), and HEK293T cells were cultured in DMEM supplemented with 10% FBS. NCI-H1048 (human SCLC) was cultured in DMEM:F12 supplemented with 0.005 mg/mL insulin, 0.01 mg/mL transferrin, 30 nmol/L sodium selenite, 10 nmol/L hydrocortisone, 10 nmol/L β -estradiol, 4.5 mmol/L L-glutamine, and 5% FBS. Penicillin-streptomycin was added to all cell culture. All cell lines were cultured at 37°C with 5% CO₂, and were periodically tested negative for *Mycoplasma* contamination. All experiments were performed within 1 month of thawing a cryopreserved vial of cells.

To make myo-inositol-deficient cell culture, customized RPMI medium without myo-inositol (Bio-Techne) was supplemented with dialyzed FBS (#26400044, Thermo Fisher Scientific) and penicillin-streptomycin. The dialyzed FBS shows 100 times less myo-inositol level compared with regular FBS, as validated by LC/MS. To measure cell growth in myo-inositol-deficient cell medium as shown in Fig. 4, cells were washed twice with PBS to remove residue cell medium, and then resuspended in myo-inositol-deficient medium with 1 million cells per mL. Cells (1 mL) were transferred into one well in 12-well plates, with three replicates for each condition. Cell number and viability were measured using automated cell counter (Invitrogen Countess) every 2 days. For each passage, 0.4×10^6 (for MOLM-13 cells) or 0.2×10^6 (for other lines) cells were passed on to the next passage. For MOLM-13, MV4-11, and U937 lines that are sensitive to myo-inositol depletion, the cell numbers dropped drastically after a second or third passage in myo-inositol-deficient medium. For these samples, 1/5 volume of the cells were passed to the next passage. We performed at least two biological replicates (each included three technical replicates) for each cell line tested. For (+MI) cell medium, 0.2 g/L (1.11 mmol/L) myo-inositol (#I5125, Sigma) was added into the myo-inositol-deficient medium.

Animals

NOD.Cg-*Prkdc*^{scid} *Il2rg*^{tm1Wjl} Tg(CMV-IL3,CSF2,KITLG)1Eav/MloYsZJ (known as NSGS) mice expressing human IL3, GM-CSF, and SCF were used for *in vivo* CRISPR screening. NOD.Cg-*Prkdc*^{scid} *Il2rg*^{tm1Wjl}/SzJ (known as NSG) mice were used for validating individual gene knockout *in vivo*. All animals were purchased from The Jackson Laboratory. Animal procedures and studies were reviewed and approved by the Institutional Animal Care and Use Committee at Cold Spring Harbor Laboratory (New York, NY).

Plasmid Construction

The lentiviral Cas9 vector (LentiV_Cas9_puro/Addgene #108100), the lentiviral sgRNA vectors (LRG2.1-GFP/Addgene #108098, LRG2.1-mCherry/Addgene #108099), the lentiviral luciferase vector (LentiV_Neo_Luc/Addgene #105621), and the lentiviral cDNA expression vector (LentiV_Neo/Addgene #108101) have been described in previous studies (42, 75). DNA oligos of the sgRNAs were cloned into LRG2.1-GFP/mCherry vectors using a *BsmBI* restriction site. The cDNAs of *ISYNA1* (#OHu18552, GenScript) and *SLC5A3* (#53836,

Addgene) were cloned into LentiV_Neo vector using the In-Fusion cloning system (Takara Bio), and FLAG tag was inserted to the C-terminus of the cDNAs. The same system was used to mutate the protospacer adjacent motif (PAM) sequences in the *SLC5A3* gene. Dual targeting vectors were generated by inserting the sgRNA1-scaffold-bU6-sgRNA2 sequence (synthesized as gene blocks by IDT) into the *BsmBI* site of the LRG2.1-GFP-P2A-BlastR vector using Gibson Assembly (NEB). The scaffold sequence for sgRNA1 was generated by replacing the stem-stem loop region of the LRG2.1 scaffold (76) with a previously described CRISPRi sgRNA scaffold (77).

Construction of a Domain-Focused sgRNA Library

The kinase sgRNA library and transcription factor sgRNA library have been described in previous studies (39, 42). The other sgRNA libraries were generated in a similar fashion. The gene lists were obtained from HUGO Gene Nomenclature Committee database (HGNC; <https://www.genenames.org/>). The domain-focused sgRNA libraries were designed by using the CSHL CRISPR sgRNA design tool (<https://crispr.cshl.edu/>), and at least six sgRNAs were selected for each gene or domain. Nontargeting sgRNAs as negative control and sgRNAs targeting essential genes as positive control were added into the pooled sgRNA library. The pooled library was synthesized on an array platform (Twist Bioscience) and then cloned into the *BsmBI* restriction site of LRG2.1-GFP vector using Gibson Assembly (NEB).

Lentivirus Transduction

Lentivirus was produced in HEK293T cells as described in the previous study (42). Briefly, 10 μ g of plasmid DNA was mixed with helper plasmids (5 μ g VSVG and 7.5 μ g psPAX2) and 1 mg/mL polyethylenimine (PEI 25000). The mix was then used to transfect HEK293T cells in a 10-cm dish. The mix was replaced by fresh cell media 6 hours after transfection, and lentivirus-containing supernatant was collected 24, 48, and 72 hours posttransfection. To infect cells with lentivirus, the target cells were mixed with the virus and 8 μ g/mL polybrene in 12-well plates, and then centrifuged at 1,700 rpm for 45 to 60 minutes. Cell media was changed 24 hours postinfection. For cells infected with LentiV_Neo or LentiV_Blast vector, G418 or blasticidin was added to the cell culture at day 3 postinfection, respectively.

Pooled In Vivo CRISPR Screening

The Cas9- and Luciferase (Luc)-expressing MOLM-13 cell line (MOLM-13-Cas9-Luc) was established by infecting MOLM-13 cells with LentiV_Cas9_puro and LentiV_Neo_Luc, and then selected with puromycin and G418. To improve *in vivo* engraftment efficiency, 5 million MOLM-13-Cas9-Luc cells were transplanted into two sublethally irradiated (2.5 Gy), 8-week-old female NSG mice through tail vein injection. The cells were collected from the bone marrow of the injected mice 10 days postinjection by flushing the tibia with PBS. The collected cells were then cultured in RPMI supplemented with 10% FBS, puromycin, and G418 to remove murine cells.

To determine virus titer for the sgRNA library, MOLM-13-Cas9-Luc cells were infected with different volumes of the virus using the spin infection technique as described above. A flow cytometer (Guava easyCyte, Millipore) was used to determine the infection rate by measuring the percentage of GFP-positive cells. The amount of virus that gave approximately 30% infection rate was used for further infection experiments.

To ensure at least 1,000-fold coverage of each sgRNA in the library, the amount of cells to be infected was calculated using the equation: cell number = sgRNA number/infection rate $\times$ 1,000. For instance, given the sgRNA number in the SLC sgRNA library was approximately 3,000, and the infection rate was approximately 30%, at least 10 million cells were used for one CRISPR screening. Ten

mice were planned to be injected, thus 1×10^6 cells were used for the initial infection.

To perform *in vivo* CRISPR screening, the pre-engraft MOLM-13-Cas9-Luc cells were infected with the library lentivirus. Three days postinfection, the infection rate was checked by flow cytometer to ensure approximately 30% GFP-positive cells. For the SLC sgRNA library, the infected wells were split into three groups. For group 1, 1×10^7 cells were spin-collected and kept frozen as P0. For group 2, 1×10^7 cells were kept growing in cell culture as *in vitro* CRISPR screening sample in parallel with *in vivo* CRISPR screening sample. During each passage, the percentage of GFP-positive cells was measured, and the cell number was calculated using the equation above to ensure at least 1,000-fold coverage of the sgRNA library. Group 3, 1×10^9 cells were collected, washed with PBS to remove any residual cell media, and then resuspended in 2 mL PBS (5×10^7 cells/mL). Two-hundred microliters of the cells were transplanted to one sublethally irradiated (2.5 Gy), 8-week-old female NSGS mouse via tail vein injection. A total of 9 mice were successfully injected. The engraftment and growth of the injected human AML cells were monitored by biofluorescence imaging (IVIS Spectrum System, Caliper Life Sciences). Images were taken 10 minutes after intraperitoneal injection of D-luciferin (50 mg/kg) every 3 days postinjection. On day 12 postinjection, the animals were euthanized. The spleen and bone marrow samples were collected. The spleen was broken down by gently crushing between the frosted sides of two glass slides and then was filtered through a cell strainer. The bone marrow was flushed from the tibia by injecting with PBS. Meanwhile, the *in vitro* samples were also spin-collected. The genomic DNA from P0, *in vitro* sample, individual spleen and bone marrow were purified using QIAamp DNA Mini kit (Qiagen). The genomic DNA was then diluted into 100 ng/ μ L. An equal aliquot of genomic DNA from individual spleen and bone marrow were pooled to make pooled spleen and pooled bone marrow samples. The *in vivo* screening for other sgRNA libraries was performed in a similar fashion, with the amount of cells and the number of animals adjusted according to the sgRNA library size to ensure at least 1,000 $\times$ coverage.

Barcoded sequencing libraries for P0, *in vitro*, 9 individual spleen, 9 individual bone marrow, pooled spleen, and pooled bone marrow samples were prepared using the same method as described in the previous study (42). The libraries were pooled and analyzed by paired-end sequencing using MiSeq (Illumina) with MiSeq Reagent Kit v3 (Illumina).

The sequencing data were debarcoded and subsequently mapped to the reference sgRNA library using a customized tool as described in the previous study (76). The screening data were analyzed using MAGeCK-MLE (78, 79). A beta score was calculated for each gene and negative control by the MAGeCK-MLE model. It is a measurement of gene selections similar to log fold change in differential expression analysis (79). Negative beta score indicated negative selection, and positive beta score indicated positive selection. To calculate beta score for the nontargeting negative controls, every six negative controls were pooled as a single target for the MAGeCK-MLE calculation. The data were visualized using the Prism 8 program.

Pooled In Vitro CRISPR Screening

Ten cell lines representing five different types of cancers were used for *in vitro* CRISPR screening of the SLC sgRNA library, including MOLM-13, HEL, NOMO1 (AML), K562 (CML), AsPC-1, MIA PaCa-2 (pancreatic cancer), NCI-H82, NCI-H1048 (SCLC), RH30, and RD (rhabdomyosarcoma). The Cas9-expressing cell lines were established by infecting the cells with LentiV_Cas9_puro, and then selected with puromycin. The virus titer for each cell line was determined individually as described above. To ensure 1,000-fold coverage of the library, at least 1×10^7 cells were initially infected with the library with approximately 30% infection rate. Day 3 postinfection, 1×10^7 cells were collected and kept frozen as P0. Cells were passed every 3 days,

and cell number and GFP% were checked every passage. Cells collected at passages 5 and 8 were saved as P5 and P8 samples. Genomic DNA extraction, sequencing library preparation, and data analysis were performed as described above.

For the genome-wide CRISPR screens in Supplementary Fig. S13, Cas9-expressing MOLM-13 and MOLM-13^{ISYNA1} cells were transduced with Human CRISPR Knockout Pooled Library (Brunello, Addgene #73178) with low multiplicity of infection (~0.3). The original library vector contains puromycin selection marker, which was replaced with blasticidin-GFP selection marker. The transduced cells were selected with blasticidin and saved as P0. The cells were passaged for another 10 doublings and saved as P10. Two biological replicates were performed. Sequencing of the sgRNAs was performed on NextSeq platform. Data analysis was performed as described above.

Competition-Based Cell Proliferation Assay

Cas9-expressing cells were infected with sgRNA cloned into LRG2.1-GFP/mCherry or dual-targeting vector. The percentage of GFP- or mCherry-positive cells was first measured at day 3 postinfection (P0) by flow cytometer (Guava easyCyte, Millipore), and then measured every 2 to 3 days at each passage until passage 5 or 6. The relative fluorescent% for each passage was calculated using the equation: $\text{relative fluorescent\%} = [\text{fluorescent\%}] / [\text{P0 fluorescent\%}] \times 100\%$. Decrease of the fluorescent% indicated the infected (gene knockout) cells were outcompeted by the uninfected (wild-type) cells.

Cell-Cycle Arrest Analysis

Cell-cycle analysis as shown in Supplementary Fig. S7 was performed according to the manufacturer's protocol (BD, FITC BrdU Flow Kit; #559619). Cells were pulsed with BrdU for 1 hour at 37°C and then were costained with 4',6-diamidino-2-phenylindole (DAPI) for DNA content measurement. Cells were finally analyzed with a BD LSRFortessa flow cytometer (BD Biosciences) and FlowJo software (TreeStar). The experiments were performed in duplicates.

Western Blot Analysis

To detect ISYNA1 expression levels in human AML cell lines, 1×10^7 cells were collected, and cell pellets were resuspended in 500 μL 1X Laemmli sample buffer (Bio-Rad). The cell lysate was then boiled for 10 minutes, separated by SDS-PAGE, followed by transfer to nitrocellulose membrane and immunoblotting with 1:500 diluted mouse monoclonal anti-ISYNA1(C-9) antibody (#sc-271830, Santa Cruz Biotechnology) and then with 1:10,000 diluted polyclonal rabbit anti-mouse HRP antibody (#P0260, Agilent Dako). For control, mouse monoclonal anti-HSC70 (B-6) antibody (#sc-7298, Santa Cruz Biotechnology) was used with 1:10,000 dilution.

To detect the recombinant ISYNA1-FLAG, 1×10^7 cells were collected 2 weeks postinfection with cDNA vectors, followed by G418 selection. Protein sample preparation and immunoblotting were the same as described above. For FLAG-tagged proteins, 1:5,000 diluted mouse monoclonal anti-FLAG antibody (Sigma) was used, followed by a secondary antibody of rabbit anti-mouse HRP antibody (#P0260, Agilent Dako) with 1:10,000 dilution.

In Vivo Validation Experiments

For the *in vivo* validation as shown in Fig. 3, pre-engraft MOLM-13-Cas9-Luc cells were infected with LRG2.1-sgROSA-GFP-BlastR or LRG2.1-sgSLC5A3-2-GFP-BlastR. At day 2 postinfection, blasticidin was added to the cell culture to select for infected cells. After 3 days of selection, the cells were collected, washed with PBS to remove residual media, and resuspended to 1×10^6 cells/mL. One-hundred microliters of cells were transplanted to one sublethally irradiated (2.5 Gy), 8-week-old female NSG mouse via tail vein injection. Five mice were injected for each sgRNA. AML development was monitored by biofluorescence imaging (IVIS Spectrum System, Caliper Life Sciences).

Images were taken 10 minutes after intraperitoneal injection of D-luciferin (50 mg/kg) on days 13, 16, and 19 postinjection.

For the *in vivo* validation as shown in Supplementary Fig. S8, pre-engraft MOLM-13-Cas9-Luc cells were infected with LRG2.1-sgROSA-GFP, LRG2.1-sgSLC5A3-1-GFP, LRG2.1-sgSLC5A3-2-GFP, LRG2.1-sgCD47-1-GFP, or LRG2.1-sgCD47-2-GFP. Excessive lentivirus was used to ensure more than 90% infection rate. The cells were then transplanted to NSG mice as described above.

Metabolomics Analysis by LC/MS

For the LC/MS analysis shown in Fig. 4, 0.5×10^6 cells were infected with LRG2.1-sgROSA-GFP-BlastR or LRG2.1-sgSLC5A3-2-GFP-BlastR in 12-well plates with six replicates. For the LC/MS analysis shown in Fig. 6, MOLM-13-Cas9 cells expressing empty vector (EV) or ISYNA1 were infected with LRG2.1-sgROSA-GFP-BlastR or LRG2.1-sgSLC5A3-2-GFP-BlastR in 12-well plates with five replicates. The infected cells were selected with blasticidin at day 2 postinfection for 3 days. For the LC/MS analysis shown in Supplementary Fig. S11–S11C, HEL-Cas9 and K562-Cas9 cells were infected with the indicated single- or dual-targeting sgRNA vectors with more than 90% infection rate, in 12-well plates with five replicates, for 5 days. The cells were then collected for sample preparation. For the LC/MS analysis shown in Supplementary Fig. S11–S11D, HEL-Cas9 cells were infected with LRG2.1-sgROSA-GFP-BlastR or LRG2.1-sgISYNA1-GFP-BlastR in 12-well plates with five replicates. The infected cells were selected with blasticidin at day 2 postinfection for 7 days. At least 95% of GFP-positive rate was achieved. The cells were then washed twice with PBS and cultured in myo-inositol-deficient medium for another 4 days before preparation for LC/MS.

To prepare cells for LC/MS, cells were quickly washed in PBS before adding 1 mL of ice-cold extraction solution (50% methanol, 30% acetonitrile, 20% H₂O) per 4×10^6 cells. The cells were then scrapped, and the suspension was snap frozen in liquid nitrogen. Samples were agitated using a Thermomixer (Eppendorf) at 1,400 rpm, 4°C for 15 minutes, followed by incubation at –80°C for 1 hour. Samples were then centrifuged at 15,000 rpm, 4°C for 10 minutes. The supernatants were further centrifuged for another 10 minutes. The supernatants were stored in autosampler vials at –80°C until analysis.

Intracellular extracts from five to six independent cell cultures were analyzed for each condition. Samples were randomized to avoid bias due to machine drift and processed blindly. LC/MS analysis was performed using a Vanquish Horizon UHPLC system coupled to a Q Exactive HF mass spectrometer (both Thermo Fisher Scientific). Sample extracts (5 μL) were injected onto a Sequant ZIC-pHILC column (150 mm $\times$ 2.1 mm, 5 μm) and a guard column (20 mm $\times$ 2.1 mm, 5 μm) from Merck Millipore kept at 45°C. The mobile phase was composed of 20 mmol/L ammonium carbonate with 0.1% ammonium hydroxide in water (solvent A) and acetonitrile (solvent B). The flow rate was set at 200 $\mu\text{L}/\text{minute}$ with the previously described gradient (80). The mass spectrometer was operated in full MS and polarity switching mode. The acquired spectra were analyzed using XCalibur Qual Browser and XCalibur Quan Browser software (Thermo Fisher Scientific) by referencing to an internal library of compounds.

Stable Isotope Tracing by LC/MS

For tracing *de novo* myo-inositol synthesis from ¹³C-labeled glucose as shown in Supplementary Fig. S14, ¹³C-labeling medium was prepared by adding ¹³C-labeled glucose (Sigma-Aldrich 389374; final concentration 2,000 mg/L) and dialyzed FBS (final concentration 10%) to glucose-free RPMI (Thermo Fisher Scientific 11879020). Cells were washed with PBS twice and seeded in 6-well plates with ¹³C-labeling medium for 48 hours. Extracts from six independent cell cultures were analyzed for each condition. Briefly, after quick

removal of the culture medium, cells were washed in PBS before adding extraction solution (50% methanol, 30% acetonitrile, 20% H₂O). After centrifugation to remove the precipitated proteins and insoluble debris, the supernatants were collected and stored in autosampler vials at -80°C until analysis. Samples were randomized to avoid bias due to machine drift and processed blindly. LC/MS analysis was performed using a Vanquish Horizon UHPLC system coupled to a Q Exactive HF mass spectrometer (both Thermo Fisher Scientific). Sample extracts were analyzed as previously described (80). The acquired spectra were analyzed using XCalibur Qual Browser and XCalibur Quan Browser software (Thermo Fisher Scientific) by referencing to an internal library of compounds. Metabolite peak areas were corrected for natural ¹³C abundance using the R package AccuCor (81). To calculate isotopolog distribution, corrected peak areas of each metabolite's isotopologs were normalized to the total metabolite pool (sum of all isotopologs of a given metabolite). Values of release of lactate were adjusted to cell density upon background subtraction.

Nanopore Sequencing

CRISPR RNA (crRNA) guides specific to the regions of interest (ROI) were designed as per recommended guidelines described in the Nanopore infosheet on Targeted, amplification-free DNA sequencing using CRISPR/Cas (Version: ECL_S1014_v1_revA_11Dec2018). The crRNA sequences are provided in Supplementary Table S2. Guides were reconstituted to 100 μmol/L using TE (pH 7.5) and pooled into an equimolar mix. For each distinct sample, four identical reactions were prepared in parallel using 5 μg gDNA each. Ribonucleoprotein complex (RNP) assembly, genomic DNA dephosphorylation, and Cas9 cleavage were performed as described in Gilpatrick and colleagues (82). Affinity-based Cas9-mediated enrichment (ACME) using Invitrogen His-Tag Dynabeads was performed to pull down Cas9-bound nontarget DNA, increasing the proportion of on-target reads in the sample (83). The resultant product was cleaned up using 1X Ampure XP beads (Beckman Coulter, #A63881), eluted in nuclease-free water, and pooled together. The ACME-enriched sample was quantified using Qubit fluorometer (Thermo Fisher Scientific) and carried forward to the adapter ligation step as described by Iyer and colleagues (83). Sequencing adapters from the Oxford Nanopore Ligation Sequencing Kit (ONT, SQK-LSK109) were ligated to the target fragments using T4 DNA ligase (NEBNext Quick Ligation Module E6056). The sample was cleaned up using 0.3X Ampure XP beads (Beckman Coulter, #A63881), washed with long-fragment buffer (LFB; ONT, SQK-LSK109), and eluted in 15 μL of elution buffer (EB; ONT, LSK109) for 30 minutes at room temperature. The resultant library was prepared for loading as described in the Cas9-mediated PCR-free enrichment protocol from ONT (Version: ENR_9084_v109_revH_04Dec2018) by adding 25 μL sequencing buffer (SQB; ONT, LSK109) and 13 μL loading beads (LB; ONT, LSK109) to 12 μL of the eluate. Each sample was run on a FLOMIN106 R9.4.1 flow cell using the GridION sequencer.

Real-time basecalling was performed with Guppy v3.2, and files were synced to our Isilon 400NL storage server for further processing on the shared CSHL high-performance computing cluster (HPCC). Nanopolish v0.13.2 (84) was used to call methylation as per the recommended workflow. Briefly, indexing was performed to match the ONT fastq read IDs with the raw signal level fast5 data. The ONT reads were then aligned to the human reference genome (UCSC hg38) using minimap2 v2.17 (85), and the resulting alignments were sorted with samtools v0.1.19 (86). Nanopolish call-methylation was then used to detect methylated bases within the targeted regions, specifically 5-methylcytosine in a CpG context. The initial output file contained the position of the CpG dinucleotide in the reference genome and the methylation call in each read. A positive value for log_{lik_ratio} was used to indicate support for methylation, using a cutoff value of 2.0. The helper script calculate_methylation.py was then used to calculate the frequency of methylation calls by genomic position.

qPCR Analysis of Indel Mutations

To analyze sgRNA editing of genomic DNA as shown in Supplementary Fig. S5A, cells were infected with LRG2.1-sgROSA-GFP-BlastR, LRG2.1-sgSLC5A3-1-GFP-BlastR, or LRG2.1-sgSLC5A3-2-GFP-BlastR. At day 2 postinfection, blasticidin was added to the cell culture to select for infected cells. After 3-day selection, the cells were collected, and genomic DNA was extracted using QIAamp DNA Mini kit (Qiagen). For qPCR, 20 to 25 ng of genomic DNA was used in each reaction with the indicated primer set. The C_t value of non-sgRNA-target primer was used to normalize the data. To analyze SLC5A3 (r-1 and r-2) expression as shown in Supplementary Fig. S6E, MOLM-13 cells were infected with the indicated cDNA constructs and were then selected with G418 for 2 weeks. Total RNA was extracted using TRIzol (Thermo Fisher Scientific), and cDNA was synthesized using qScript cDNA SuperMix (Quanta). All qPCR experiments were performed using Power SYBR Green PCR Master Mix (Thermo Fisher Scientific) on a QuantStudio Flex Real-time PCR machine (Thermo Fisher Scientific).

AML Patient Data

TCGA PanCancer AML patient data, including gene expression and mutation, were downloaded via cBioPortal (<https://www.cbioportal.org/>; refs. 87-89). The same patients' DNA methylation data were downloaded via NCI Genomic Data Commons (<https://gdc.cancer.gov/>; ref. 55). The AML (*n* = 119) and normal (*n* = 22) individual data from Glass and colleagues were downloaded from Gene Expression Omnibus (GEO; accession number GSE98350; ref. 56). After filtering and normalizing by coverage, a methylBase object containing the methylation information and locations of cytosines that were present in at least five samples per condition (meth.min = 5) was generated using MethylKit (version 1.9.3) and R statistical software (version 3.5.1). Percent methylation for each CG for each donor was calculated using the MethylKit "percMethylation" function. The Bedtools intersect function was used to determine overlap with CpG islands from hg19.

Chromatin Immunoprecipitation Sequencing Data Analysis

For chromatin immunoprecipitation sequencing (ChIP-seq) analysis shown in Supplementary Fig. S15A, raw reads were obtained from public GEO datasets, MOLM-13 (GSE63782/GSM1652920), K562 (GSE63782/GSM1652918; ref. 90), MV-411 (GSE79899/GSM2108039), THP-1 (GSE79899/GSM2108046; ref. 91), and HEL (GSE109492/GSM2944364; ref. 42), and mapped to the human genome (hg19) using Bowtie2 software with sensitive settings (92). Duplicate reads were removed prior to peak calling using MACS2 software using 5% FDR cutoff and broad peak option (93). Sequencing depth normalized ChIP-seq pileup tracks were visualized using the Integrative Genomics Viewer genome browser (94).

Statistical Analysis

Error bars represent the mean plus or minus SEM, and *n* refers to the number of biological repeats. Statistical significance was evaluated by *P* value using GraphPad Prism software or Scipy package from Python 3.8.5, as indicated in the figure legends. For Kaplan-Meier survival curves, the log-rank (Mantel-Cox) test was used to estimate median overall survival and statistical significance.

Authors' Disclosures

S.V. Iyer reports travel bursaries from Oxford Nanopore Technologies to cover expenses (travel, registration fees, food, etc.) incurred at conferences and presented talks/posters in the past. O. Klingbeil reports other support from Deutsche Forschungsgemeinschaft during the conduct of the study. W.R. McCombie reports grants from the NIH during the conduct of the study, as well as nonfinancial support from Pacific Bioscience, personal fees and other support from Orion

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Authors' Contributions

Y. Wei: Conceptualization, investigation, methodology, writing—original draft, writing—review and editing. **Y.-H. Huang:** Investigation. **D.S. Skopelitis:** Investigation. **S.V. Iyer:** Investigation, methodology. **A.S.H. Costa:** Investigation, methodology. **Z. Yang:** Investigation. **M. Kramer:** Investigation. **E.R. Adelman:** Investigation. **O. Klingbeil:** Investigation, methodology. **O.E. Demerdash:** Investigation. **S.A. Polyanskaya:** Investigation. **K. Chang:** Investigation. **S. Goodwin:** Investigation. **E. Hodges:** Supervision, investigation. **W.R. McCombie:** Supervision, investigation. **M.E. Figueroa:** Supervision, investigation. **C.R. Vakoc:** Conceptualization, supervision, writing—review and editing.

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